```
GNU nano 8.7                                                    luc.c
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int frames[3];
    int time[3];
    int i,j,found=0,least,pos;
    int faults=0,hits=0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        found = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                hits++;
                time[j] = i;
                found = 1;
                break;
            }
        }

        if(!found) {
            least = time[0];
            pos = 0;

            for(j=0;j<3;j++) {
                if(frames[j] == -1) {
                    pos = j;
                    break;
                }
                if(time[j] < least) {
                    least = time[j];
                    pos = j;
                }

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy
```

```
GNU nano 8.7                                                    luc.c
                hits++;
                time[j] = i;
                found = 1;
                break;
            }
        }

        if(!found) {
            least = time[0];
            pos = 0;

            for(j=0;j<3;j++) {
                if(frames[j] == -1) {
                    pos = j;
                    break;
                }
                if(time[j] < least) {
                    least = time[j];
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            time[pos] = i;
            faults++;
        }
    }

    printf("\nLRU Page Replacement\n");
    printf("Page Faults = %d\n", faults);
    printf("Page Hits = %d\n", hits);

    printf("Hit Percentage = %.2f%%\n", (float)hits/n*100);
    printf("Fault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy
```

```
MAYUR LAP MSYS ~
$ nano luc.c

MAYUR LAP MSYS ~
$ gcc luc.c -o luc

MAYUR LAP MSYS ~
$ ./luc

LRU Page Replacement
Page Faults = 10
Page Hits = 2
Hit Percentage = 16.67%
Fault Percentage = 83.33%

MAYUR LAP MSYS ~
$
```